

SIMPLE AI GUIDE

Smart Home Assistant Robot

Explained clearly using the **PEAS Framework**: Performance, Environment,
Actuators, and Sensors.

1. Performance & Environment



Performance (P) — How well it works

- ✓ **Safety First:** Never bump into humans, pets, or furniture.
- ✓ **Fast & Useful:** Complete home chores quickly and correctly.
- ✓ **Smart Battery:** Go to charger automatically before battery dies.
- ✓ **Quiet & Private:** Work quietly and keep personal family data safe.



Environment (E) — Where it works

- ✓ **Home Spaces:** Living rooms, kitchens, stairs, and carpets.
- ✓ **Moving Obstacles:** People walking around, playing pets, toys on floor.
- ✓ **Changing Lighting:** Bright rooms during day, dark rooms at night.



2. Actuators & Sensors



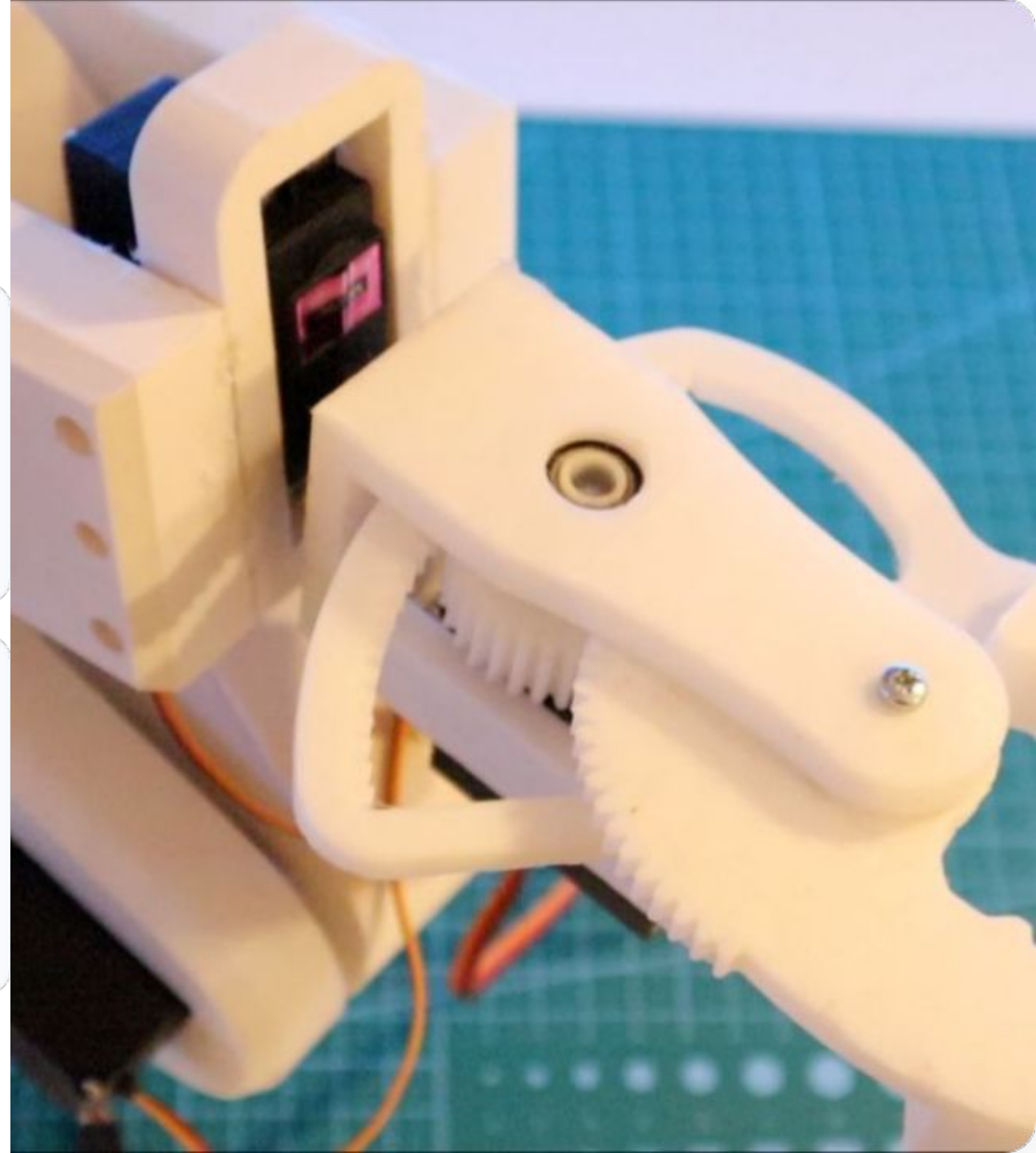
Actuators (A) — How it takes action

- ✓ **Wheels / Legs:** To move smoothly across different floors.
- ✓ **Robotic Arm:** To pick up items, open doors, or deliver food.
- ✓ **Speaker & Screen:** To talk back and show helpful information.



Sensors (S) — How it sees and hears

- ✓ **Cameras & LiDAR:** To see obstacles and build a map of the house.
- ✓ **Microphones:** To hear voice commands from anywhere in the room.
- ✓ **Touch & Cliff Sensors:** To prevent falling down stairs or crashing.



3. PEAS Summary at a Glance

PEAS Letter	Meaning	Simple Example
P 	Performance	No collisions, fast response time, energy efficiency, high safety.
E 	Environment	Indoor home, family members, pets, stairs, furniture, dark/lit rooms.
A 	Actuators	Wheels for moving, arm for grabbing objects, speakers for talking.
S 	Sensors	Cameras for vision, microphones for hearing, LiDAR for distance.

Thank You!

Any questions about how smart assistant robots work?



Artificial Intelligence



Robotics Made Simple

Image Sources



https://media.easy-peasy.ai/a30799d5-64e1-4893-adf6-88f4efda7f44/29fc86f3-19ec-44ec-8b3e-43d7ac1d6759_medium.webp

Source: easy-peasy.ai



<https://smartbuilds.io/wp-content/uploads/2021/01/robotic-arm-top-close-up.jpg>

Source: smartbuilds.io